QCrypt 2019 Schedule (subject to changes)

Monday, 26 August 2019

Monday morning

Tutorial
Norbert Lütkenhaus: QKD implementation security

Invited
Ronald Hanson: Quantum repeaters and quantum networks

Contributed

Gayane Vardoyan, Saikat Guha, Philippe Nain and Don Towsley.
On the capacity region of bipartite and tripartite entanglement switching and key distribution

Daniel Llewellyn, Caterina Vigliar, Benjamin Slater, Beatrice Da Lio, Stefano Paesani, Jorge Barreto, Dondu Sahin, Massimo Borghi, John G. Rarity, Leif K. Oxenløwe, Karsten Rottwitt, Jianwei Wang, Yunhong Ding, Mark G. Thompson and Davide Bacco.
High-dimensional chip-to-chip entanglement distribution through multicore fibre

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Monday afternoon

Invited
Ji-Gang Ren: QKD based on satellite-ground entanglement distribution

Contributed

Marco Avesani, Luca Calderaro, Matteo Schiavon, Costantino Agnesi, Alberto Santamato, Andrea Stanco, Mujtaba Zahidy, Alessia Scriminich, Giulio Foletto, Giampiero Contestabile, Marco Chiesa, Alessandro Nottola, Davide Rotta, Stefano Tirelli, Massimo Artiglia, Alberto Montanaro, Marco Romagnoli, Vito Sorianello, Daniele Dequal, Giuseppe Bianco, Claudia Facchinetti, Alberto Tuozzi, Francesco Vedovato, Giuseppe Vallone and Paolo Villoresi.
QCoSOne: A chip-based prototype for daylight free-space QKD at telecom wavelength for future satellite optical payloads

Henry Semenenko, Philip Sibson, Andy Hart, Mark Thompson and Chris Erven.
Chip-based measurement-device-independent quantum key distribution

Tobias Eriksson, Takuya Hirano, Benjamin Puttnam, Georg Rademacher, Ruben Luís, Mikio Fujiwara, Ryo Namiki, Yoshinari Awaji, Masahiro Takeoka, Naoya Wada and Masahide Sasaki.
Continuous variable quantum key distribution multiplexed with high throughput coherent channels

Break

Ignatius William Primaatmaja, Emilien Lavie, Koon Tong Goh, Chao Wang and Charles Ci Wen Lim.
Almost-tight and versatile security analysis of measurement-device-independent quantum key distribution

Niraj Kumar, Iordanis Kerenidis and Eleni Diamanti.
Experimental demonstration of quantum advantage for one-way communication complexity with application in construction of robust quantum money

Marie-Christine Roehsner, Joshua Kettlewell, Tiago Batalhao, Joseph Fitzsimons and Philip Walther.
Quantum advantage for probabilistic one-time programs

Tuesday, 27 August 2019

Tuesday morning

Tutorial
Rotem Arnon-Friedman: Tutorial on DI-QKD and DI-QRNG, discussing security proofs and practical challenges

Invited
Eneet Kaur: Fundamental limits on key rates in DI-QKD

Contributed

René Schwonnek, Ernest Y.-Z. Tan, Ramona Wolf, Koon Tong Goh and Charles C.-W. Lim.
A numerical method for computing reliable secret key rates for device-independent quantum key distribution

Alex Bredariol Grilo.
A simple protocol for verifiable delegation of quantum computation in one round

Rotem Arnon-Friedman and Jean-Daniel Bancal.
Device-independent certification of one-shot distillable entanglement

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Tuesday afternoon

Free afternoon

Wednesday, 28 August 2019

Wednesday morning

Tutorial
Fang Song: Tutorial on quantum zero knowledge

Invited
Mark L. Zhandry: Quantum techniques in post-quantum crypto

Contributed

Thomas Vidick and Tina Zhang.
Classical zero-knowledge arguments for quantum computations

Alex Bredariol Grilo, William Slofstra and Henry Yuen.
Perfect zero knowledge for quantum multiprover interactive proofs

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Wednesday afternoon

Invited
Stefanie Barz: Implementation vulnerabilities in general quantum cryptography

Contributed

Kento Maeda, Toshihiko Sasaki and Masato Koashi.
Operator dominance method: a simple monitoring scheme of a TF-type QKD in finite-size regime

Yang Liu, Zong-Wen Yu, Weijun Zhang, Jian-Yu Guan, Jiu-Peng Chen, Chi Zhang, Xiao-Long Hu, Hao Li, Teng-Yun Chen, Lixing You, Zhen Wang, Xiang-Bin Wang, Qiang Zhang and Jian-Wei Pan.
Experimental Twin-field quantum key distribution through sending-or-not-sending

Mariella Minder, Mirko Pittaluga, George L. Roberts, Marco Lucamarini, James F. Dynes, Zhiliang Yuan and Andrew J. Shields.
Experimental twin field quantum key distribution beyond the repeaterless secret key capacity bound

Xiaoqing Zhong, Jianyong Hu, Marcos Curty, Li Qian and Hoi-Kwong Lo.
Proof-of-principle experimental demonstration of twin-field type quantum key distribution

Break

Anne Broadbent and Sébastien Lord.
Uncloneable quantum encryption via oracles

Myrto Arapinis, Mahshid Delavar, Mina Doosti and Elham Kashefi.
Security analysis of quantum physical unclonable functions

Norbert Lütkenhaus, Ashutosh Marwah and Dave Touchette.
Erasable bit commitment from temporary quantum trust

Thursday, 29 August 2019

Thursday Morning

Tutorial
Manfred Lochter: Standards for true random / certification of QRNG and QKD

Invited
Zvika Brakerski: Certifiable randomness from a single quantum device

Contributed

Yanbao Zhang, Honghao Fu, Krister Shalm, Joshua Bienfang, Martin Stevens, Michael Mazurek, Sae Woo Nam, Carlos Abellan, Waldimar Amaya, Morgan Mitchell, Carl Miller, Alan Mink and Emanuel Knill.
Efficient randomness certification by quantum probability estimation

Davide Rusca, Thomas van Himbeeck, Anthony Martin, Jonatan Bohr Brask, Hamid Tebyanian, Stefano Pironio, Nicolas Brunner and Hugo Zbinden.
Fast and practical implementation of self-testing QRNG based on an energy bound
merged with
Thomas Van Himbeeck and Stefano Pironio.
Correlations and randomness generation based on an energy constraint

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Thursday afternoon

Invited
Zhiliang Yuan: 10~Mb/s quantum key distribution

Industry session (separate schedule)

Friday, 30 August 2019

Friday Morning

Invited
Gorjan Alagic: Signing/encrypting quantum states

Contributed

Jelle Don, Serge Fehr, Christian Majenz and Christian Schaffner.
Security of the Fiat-Shamir transformation in the quantum random-oracle model

Fabio Banfi, Ueli Maurer, Christopher Portmann and Jiamin Zhu.
Composable and finite computational security of quantum message transmission

Jan Czajkowski, Christian Majenz, Christian Schaffner and Sebastian Zur.
Quantum lazy sampling and game-playing proofs for quantum indifferentiability

Alexandru Gheorghiu and Thomas Vidick.
Computationally-secure and composable remote state preparation

Christian Majenz, Christian Schaffner and Jeroen van Wier.
Non-malleability for quantum public-key encryption